



Rounding to any number of decimal places

Task A: Rounding to a given number of decimal places

1) Complete the following table:

Number	to 1 d.p.	to 2 d.p.	to 3 d.p.
14.25342	14.3	14.25	14.253
765.3246	765.3	765.32	765.325
2035.83716	2035.8	2035.84	2035.837
7.658192	7.7	7.66	7.658
82.82926	82.8	82.83	82.829
0.63492	0.6	0.63	0.635
0.08739	0.1	0.09	0.087
0.2995	0.3	0.30	0.300

2) Using the clues, find the numbers being described:

a) Clue 1 – I am between 2 and 3.

Clue 2 – I have 3 digits after the decimal point.

Clue 3 – When rounded to 2 decimal places I am 2.88.

Clue 4 – My hundredths digit is greater than my thousandths digit.

Clue 5 – The difference between my tenths and thousandths digits is 2.

I am 2.876

b) Clue 1 – I am between 1 and 2.

Clue 2 – I have 3 digits after the decimal point.

Clue 3 – When rounded to 1 decimal place I am 1.9.

Clue 4 – When rounded to 2 decimal places I am 1.90.

Clue 5 – The sum of my tenths and thousandths column is 13.

I am 1.904 or 1.895

2) Using the clues find the numbers being described:

c) Clue 1 – I am between 0 and 1.

Clue 2 – I have 3 digits after the decimal point.

Clue 3 – When rounded to 1 decimal place I am 0.1.

Clue 4 – When rounded to 2 decimal places I am 0.08.

Clue 5 – My hundredths column is double my thousandths column.

I am 0.084

Task B: Rounding in a context

1) 3 friends go out for lunch and spend £25. They decide that there are 3 different ways they can pay.

a) Write down the pros and cons of the different ways.

i) Round the cost to the nearest £10 and split the bill.

$£25 \approx £30$ (to the nearest ten)

$£30 \div 3 = £10$

They pay £10 each.

They cover the cost of lunch but pay £5 more than they needed to.

ii) Split the bill between the 3 of them, round this to the nearest pound and each pay that amount.

$£25 \div 3 = £8$ (to the nearest pound)

$£8 \times 3 = £24$

It is easy to work out but they are £1 short on the cost of lunch.

iii) Split the bill between the 3 of them, round to 2 decimal places and each pay that amount.

$£25 \div 3 = £8.33$ (to 2 d.p.)

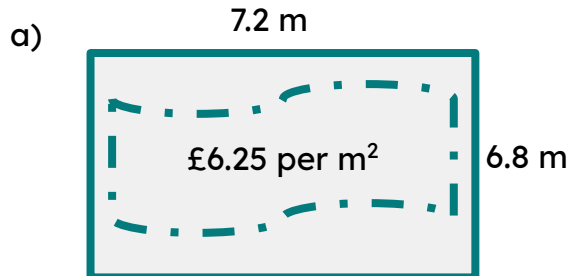
$£8 \times 3.33 = £24.99$

They each pay the same but they are 1 p short on the cost of lunch.

b) What should they do to pay for the lunch?

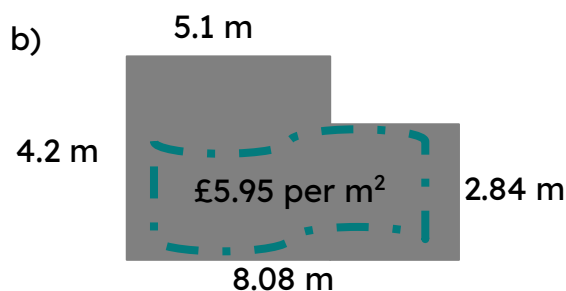
They should divide the cost by 3 but round up to the nearest penny.

2) Work out the cost of carpeting the following:



$$7.2 \times 6.8 = 49.0 \text{ m}^2$$

$$49.0 \times 6.25 = \text{£}306.25$$



$$8.08 \times 4.2 - 2.98 \times 1.36 = 29.88 \text{ m}^2$$

$$29.88 \times 5.95 = \text{£}177.79$$

3) Lucas, Aisha and Jacob take part in 800 m on sports day. The teacher recorded their times to the nearest minute.

Here are their times:

Lucas

Aisha

Jacob

3 minutes

3 minutes

3 minutes

a) Was rounding to the nearest minute a suitable degree of accuracy?

No, because the times were all the same.

b) The teacher then decides to record their times to the nearest hundredth of a second. Is this an appropriate degree of accuracy?

Yes, this should definitely separate the pupils.

Here are their exact times :

Lucas

Aisha

Jacob

00:03.08

00:03.20

00:02.58

c) Was it necessary to record the times to the nearest 100th of a second?

It was not necessary, as you could tell from the tenths of a second.